A Differentiated Tariff Strategy to Strengthen U.S. Access to Clean Energy Technologies

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Executive Summary

The recent expansion of U.S. tariffs on Chinese imports poses a severe threat to access to clean energy supplies in the United States. The Chinese imports this analysis focuses on include solar panels and batteries, which play a crucial role in the U.S. meeting its climate and energy goals. The reciprocal tariffs announced by China add layers of uncertainty and threaten the resurgence of fossil fuel energy expansion. These tariffs raise costs for U.S. solar deployment and electric vehicle (EV) battery production. (Yellen & Sagatom, 2025)

The purpose of this analysis is to determine which tariff strategy best protects U.S. trade interests and effectively advances the nation's clean energy goals. Specifically, we evaluate tariff levels that:

1) Minimize technology costs (Levelized Cost of Electricity and EV affordability),
2) Strengthen U.S. and allied supply chains for solar and batteries, and
3) Reduce strategic dependence on China where it poses national or economic security risks. (Yellen & Sagatom, 2025)

By integrating market data, tariff impacts, stakeholder priorities, and multi-criteria decision analysis, the goal is to identify a tariff structure that maximizes U.S. energy security and clean-energy deployment while supporting domestic industry growth. The recommendation proposed is that the U.S. adopt a differentiated country-specific tariff strategy. This includes maintaining the current 50% tariff on Chinese solar-industry imports and keeping low tariff rates for non-Chinese partners importing solar panels. In parallel, it is suggested that tariffs on Chinese EV batteries be reduced from 25% to 7.5% to relieve cost bottlenecks and accelerate U.S. EV adoption.

Context of Tariffs in the Solar and Battery Sectors

In 2018, under the first Trump administration, the U.S. imposed tariffs on a wide range of Chinese goods following an investigation under Section 301 of the U.S. Trade Act. This provision authorizes the U.S. Trade Representative to investigate foreign trade practices and impose remedies—such as tariffs—on those deemed unfair or discriminatory to protect the domestic industry. In the battery sector, concerns about intellectual-property theft and the unfair subsidization of key technologies in China led the United States to impose a 10% tariff on Chinese lithium-ion batteries. (USTR, n.d.) As trade tensions escalated in May 2019, this tariff was increased to 25%

and has remained in place through the Biden administration, despite pressure from automakers and clean-energy advocates to reduce it. (LaMonica, 2020)

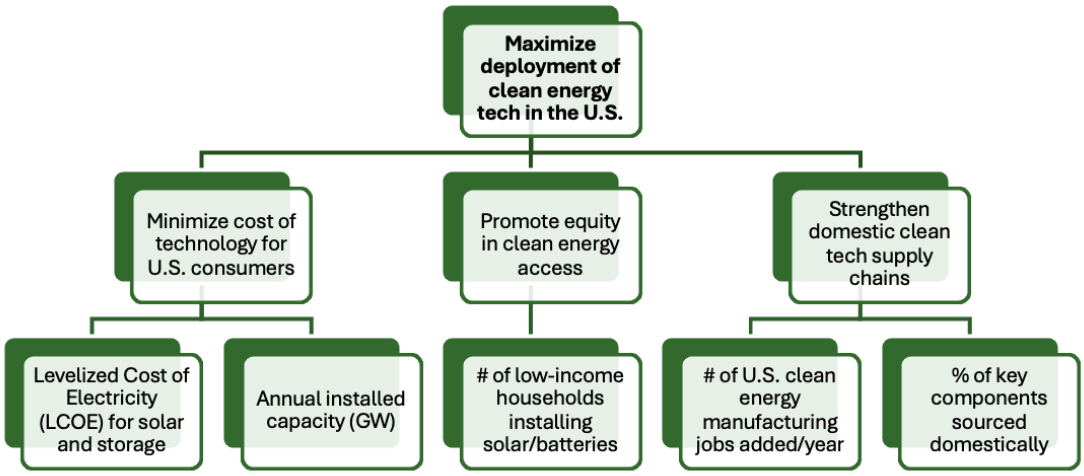
The solar industry followed a very similar pattern, with the Trump administration imposing Section 201 tariffs in 2018 on imported solar panels and modules from China. The initial plan was for these tariffs to begin at 30% and decline by 5% each year as U.S. manufacturing became more competitive. (CRS, 2019) However, in 2022, the Energy Information Administration (EIA) found that roughly 88% of U.S. solar panels were still imported, as Chinese manufacturers shifted production to countries such as Vietnam, Malaysia, and Thailand to circumvent tariffs. (EIA, n.d.) Therefore, in 2024, the Trump administration imposed its most stringent solar tariffs to date, ranging from 14% to 3,500%. The highest rates were applied to countries deemed connected to Chinese dumping, while more favorable rates were given to those that cooperated with U.S. trade investigations. (Marshall, 2025)

The key stakeholders in this decision include U.S. solar manufacturers, EV and battery manufacturers, utilities, low-income households, environmental organizations, and the federal government. U.S. solar manufacturers generally favor maintaining high tariffs on Chinese solar modules, as these measures protect emerging domestic capacity and have already stimulated significant investment following the IRA. EV manufacturers and grid-storage developers, however, largely oppose high tariffs on batteries, warning that elevated costs will slow EV adoption, delay storage deployment, and weaken U.S. competitiveness. Utilities and large energy buyers are aligned with this view, as high battery costs increase system-wide LCOE and raise the cost of meeting clean-energy targets. Low-income households and equity-focused organizations similarly support reducing battery tariffs because EV and residential storage affordability are highly sensitive to battery prices. The federal government is internally split: trade and national-security agencies prioritize reducing dependence on China by imposing tariffs, whereas energy and climate agencies emphasize affordable deployment and rapid decarbonization. Environmental NGOs generally support a differentiated tariff approach that ultimately protects U.S. consumers and accelerates climate change mitigation.

Decision Tree and Metrics

This decision tree outlines the project's core focus and guides the direction of the recommendation process. The overarching goal is to maximize the deployment of clean-energy technologies in the United States, and to reach that goal, the policy framework must address three key sub-objectives. First, it is essential to minimize the cost of this technology for U.S. consumers to make

clean-energy options more attractive and competitive at both the individual and corporate levels. Next, promote equity in clean-energy access. Since these technologies are often adopted first by higher-income individuals and households, low-income and marginalized communities are frequently left behind. This disparity is driven not only by high upfront technology costs, but also by state incentive structures that often reward early adopters—who are typically wealthier—thereby reinforcing existing inequities. For instance, the New York "NY-Sun" solar incentive program offers long-term production-based incentives in sequential 'blocks.' Once a given megawatt target is reached, the program moves to the next block, which offers lower incentive payments. (DSIRE, n.d.) To truly maximize deployment, it's essential that access expands regardless of income or location. The last sub-goal is to strengthen domestic clean-technology manufacturing to better insulate the sector from foreign conflicts and the supply-chain disruptions that follow.



The key performance indicators are the quantitative metrics listed at the bottom of the decision tree, which measure progress towards each sub-goal.

1) The levelized cost of electricity (LCOE) for solar and storage, as well as the annual installed capacity of clean-energy technologies, are used to assess whether the policy is effectively minimizing clean-technology costs.
2) Equity in clean-energy access is measured by the number of low-income households installing solar systems or purchasing electric vehicles.
3) Lastly, the strength of the domestic clean-technology supply chain is measured by the annual number of jobs added in the clean-energy manufacturing sector and by the share of key technology components sourced domestically.

Data Insights Shaping U.S. Clean Energy Strategy

1) U.S. Import Dependence on Chinese Clean Technology

By 2030, the global battery market is projected to more than double in value, reaching approximately $332 billion, as batteries become essential components across a wide range of technologies from AI infrastructure and data centers to electric vehicles and grid-scale storage. (Yellen & Sagatom, 2025) Today, China controls roughly 75% of global battery production and much of the processing capacity for critical minerals such as lithium, nickel, cobalt, and graphite. (Sutton et al., 2025). Given this level of concentration, a rapid or aggressive decoupling from China would create severe supply shortages, raise technology costs across multiple sectors, and slow U.S. clean-energy deployment. The economic effects of these tariff rates can be significant, as in the first four months of 2024, the U.S. imported $4 billion worth of lithium-ion batteries, underscoring how deeply integrated Chinese supply chains remain in the U.S. economy. (Gordan, 2025).

Ultimately, the effects of these tariffs are felt by consumers who see electricity prices rise and by domestic manufacturers who face increasing manufacturing and storage costs. (Gordan, 2025). The impacts of these tariffs are already visible with postponed projects, cancellation of billion-dollar projects, and poor forecasts. (Temple, 2025) Of the 133,000 jobs created in the battery sector by IRA regulations, 6,000 have been lost. (Temple, 2025) For the next five to seven years, there is no cost-effective alternative to battery storage for meeting domestic electricity demand without Chinese-imported cells. (Yellen & Sagatom, 2025)

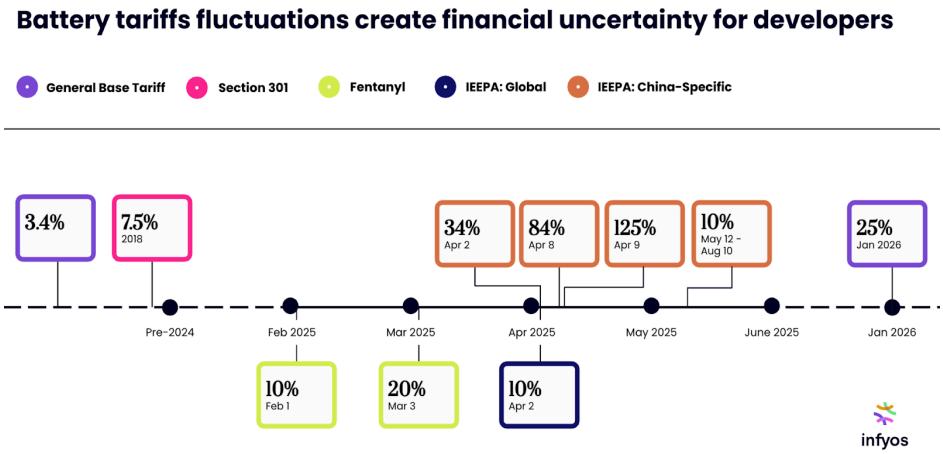


Image: How Battery Projects Can Survive Tariff Turmoil?

The solar industry shows promising results, with solar module manufacturing increasing by a factor of 5 since the passage of the IRA. This growth, coupled with U.S. policies to diversify imports of solar cells, ensures that the solar industry is less affected by tariff increases than the battery industry. (Larson, 2025) Only 4% of China's total exports of solar power go to the U.S., which means the U.S. can maintain a relatively high tariff on solar modules without risking major retaliation or supply instability. (Larson, 2025). Several Southeast Asian nations, such as Vietnam, Indonesia, and Malaysia, supply almost 88% of U.S. solar imports. As a result of federal manufacturing incentives, over 70 manufacturing facilities are active, and 47 new ones are under construction. (Kennedy, n.d.) According to the SEIA, these planned facilities are capable of meeting the U.S. solar energy demand (Kennedy, n.d.). As a result, the solar sector is structurally less vulnerable to tariff-induced cost increases compared to the battery sector, which remains deeply dependent on Chinese cell and mineral processing capacity.

Potential Policy Frameworks

There are three trade policy pathways the U.S. government could pursue, each offering a different approach to balancing affordability, equity, and domestic supply-chain resilience in the solar and lithium-ion battery sectors.

a. Free Trade Agreement

The first alternative is a free-trade approach, which would eliminate—or substantially reduce—tariffs on Chinese solar products and electric-vehicle battery components. Tariffs on imported solar photovoltaic (PV) modules and cells have caused roughly 10.5 GW of lost solar deployment. (SEIA, n.d.) As tariffs raise prices, they reduce demand, making solar less affordable for many consumers and delaying or cancelling clean-energy projects. By reducing tariffs, the United States could foster greater cooperation with China, allowing each country to specialize according to its comparative advantage. This would reduce production inefficiencies and lower prices for U.S. consumers. With that said, lower prices would make it harder for U.S. producers to enter the solar and battery markets, as they would struggle to cover their high costs at this price point. Therefore, this plan must be accompanied by extensive federal subsidies for both solar and battery producers.

b. Differentiated Tariff Strategy

A second policy approach is to implement different tariff levels for the solar and battery sectors to maximize domestic clean energy deployment without sacrificing U.S. manufacturing. Specifically, the U.S. would keep the 50% tariff on Chinese solar products, but lower the tariff

on Chinese EV batteries to 7.5%. This strategy has two primary implications. First, it will support domestic manufacturing where U.S. consumers are primed to endure slightly higher costs in the solar industry. Second, it reduces the cost of electric vehicles, making them more affordable for consumers who are highly sensitive to price changes, thereby enabling greater EV adoption. For instance, a 10% decrease in electric vehicle prices is associated with roughly a 21% increase in adoption. (Muehlegger & Rapson, n.d.)

c. High Tariffs on Both Sectors

The final policy option is to raise tariffs on both sectors, above the current 50% rate for solar products and the 25% rate for lithium-ion batteries. This would promote an autarkic shift toward severing trade relations with China and relying almost entirely on domestic manufacturing and non-Chinese suppliers. This option would sharply raise technology costs. In fact, in an economic-impact analysis of Section 301 tariffs, the U.S. International Trade Commission found that tariff increases are passed almost entirely on to U.S. consumers. For example, a 25% tariff resulted in roughly a 25% increase in import prices. (USITC, 2023) It would take years for the U.S. to build out sufficient manufacturing capacity to realistically cover most U.S. demand in a given cleantech sector. Predictive modeling shows it would take more than 6 years to build a single battery plant. (Porsche Consulting, 2024) As tariffs drive up the cost of imported technologies, federal manufacturing incentives become less necessary because domestic firms would face far less price competition and could enter the market more easily.

Utility Matrix Modeling

To assess which alternative offers the strongest overall recommendation, each option is evaluated against key stakeholders' preferences to identify the one that delivers the highest utility across those groups. This process used a multi-attribute utility theory approach, in which the utility of all five indices of performance was estimated for each of the six key stakeholders under each policy alternative. The IP with the highest stakeholder preference was assigned a value of 1, while the lowest was assigned a value of 0. The intermediate values were then determined using the principle of indifference—specifically, the probability at which a stakeholder would be indifferent between choosing the intermediate option versus taking a gamble between the highest and lowest-valued options. Lastly, weights were assigned to each IP to reflect the importance of each index to that stakeholder group. This analysis found that Alternative 2—the differentiated tariff strategy—achieves both the highest total utility and the

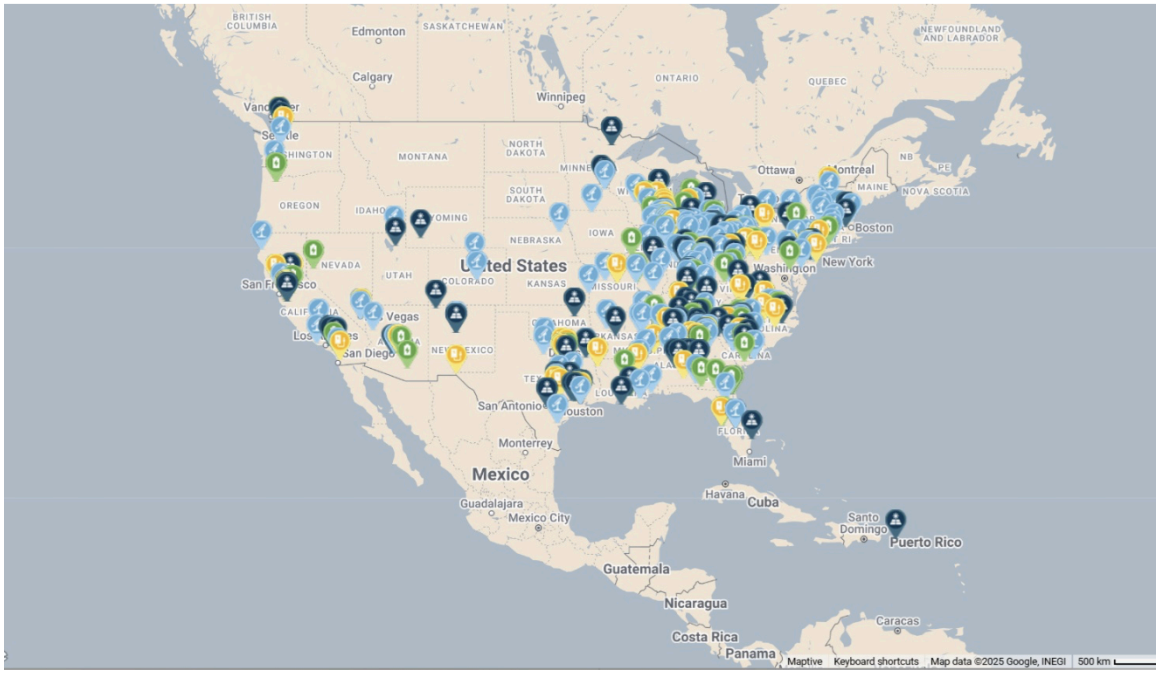
highest average utility.

					Alternative 1	Alternative 2	Alternative 3
		U.S. Federal Government					
Weights							
0.7			**IP 1**		0.55	1.00	0.00
0.7			**IP 2**		0.40	1.00	0.00
0.5			**IP 3**		1.00	0.60	0.00
0.6			**IP 4**		0.00	1.00	0.70
0.8			**IP 5**		0.00	0.25	1.00
			Sum		1.17	2.50	1.22

Recommendation: Targeted Tariff Reform to Strengthen U.S. Clean Energy Transition

Based on the analysis, the recommendation is for the U.S. to adopt a differentiated, country-specific tariff strategy. This means that the U.S. should maintain the current 50% Section 301 tariff on Chinese solar cells and modules but reduce the Section 301 tariff on lithium-ion EV batteries and battery parts to 7.5%, while continuing to expand domestic tax credits, grants, and lowered tariff rates for U.S. clean-tech manufacturers and friend-shoring country partners.

The differentiated tariff recommendation follows directly from sector-specific evidence on cost, dependency, and alternative suppliers. Recent United States Trade Representative (USTR) actions raised tariffs on Chinese solar cells and modules to 50% to protect U.S. manufacturing. In the solar industry, these tariff policies are not new, and for over a decade, resilient solar supply chains have been a focus of U.S. policy. Although the new tariff rates have pitted key stakeholders against one another, several domestic manufacturers, including Auxin Solar, SolarWorld, and Suniva, have applauded these tariffs as a step in the right direction toward tackling China's unfair trade practices and dumping. In solar, this higher tariff is also partially justified by the rapid growth of U.S. module capacity: domestic manufacturers have already met a 50 GW module production target, and the U.S. is now the 3rd largest solar module producer globally. (Kennedy, n.d.) Thanks to its diversification policies dating back to 2012, the U.S. also imports cells and wafers largely from non-Chinese suppliers in Malaysia, South Korea, and Thailand. Maintaining the 50% solar tariff, while leveraging allied supply, therefore reinforces an already successful reshoring strategy without fully cutting the U.S. off from necessary inputs. (Larson, 2025)



Map showing the location of solar manufacturing facilities in the U.S.
Image: SEIA

Conversely, the battery industry remains structurally dependent on China, as Chinese battery prices are among the lowest globally, and there is no alternative production capacity to replace these imports. The current 25% tariff on EV batteries threatens to cripple key downstream industries such as automotive and consumer electronics. These tariffs risk higher costs and slower deployment of grid-scale storage facilities and EVs. A report by Princeton's Zero's lab estimates that these tariffs could result in the cancellation of 100% of planned battery manufacturing facilities and the shutdown of 72% of currently operating plants (Michelle Lewis, 2025). Rolling EV battery tariffs back from 25% to 7.5% aligns with that evidence: it eases a key cost bottleneck in a sector where the U.S. has limited short-run substitutes, while still preserving some leverage.

Wood Mackenzie's analysis of construction cost changes in two tariff scenarios shows the dramatic impact on battery storage. Image: Wood Mackenzie

Finally, the recommendation is that the U.S. use tools such as country-specific tariff rate quotas to provide trusted partner countries with reliable, low-tariff access while limiting Chinese circumvention. This approach balances trade leverage with the need for affordable, secure access to clean technologies. Our recommendation directly supports three goals, which include:

1) Strengthening supply chains and U.S. jobs: Keeping solar tariffs high strengthens the competitive position of emerging U.S. solar producers while preventing a renewed flood of low-cost Chinese panels into the market. At the same time, lowering battery tariffs ensures the U.S. doesn't stall progress in a sector where domestic capacity isn't yet sufficient to meet demand. The result is a more resilient clean-tech supply chain in the U.S. that boosts job growth across both industries.
2) Promoting Equity: This tariff structure ensures that high-battery tariffs do not disproportionately burden low-income households and makes home batteries more affordable and accessible, resulting in wider EV adoption and lower energy costs for these households.
3) Minimizing Technology Cost: The tariff rollback on EV batteries reduces the cost of lithium-ion cells, and as battery costs significantly drive up storage and retail prices, this rollback alleviates cost pressures across the whole system. (Michelle Lewis, 2025)

The Key Actions to Implement a Differentiated Tariff Strategy Include:

- Continuing the 50% Section 301 tariff on solar cells and returning EV battery tariffs to their pre-2024 level (7.5%) to mitigate deployment bottlenecks in storage.
- Expanding the IRA tax manufacturing credits (45X), which offers per unit tax credits for manufacturing components such as solar cells and wafers.
- Expanding IRA production credits (45/48), which includes a 30% investment tax credit for building and expanding clean technology manufacturing facilities
- Allow lower tariff access for cathode precursors and battery packs sourced from allies such as South Korea and Japan to reduce reliance on China
- Deepen bilateral and multilateral cooperation by accelerating friend-shoring agreements for mineral security partnerships that support supply chain diversification
- Create joint manufacturing zones with key trading partners to share risks and lower capital costs. A good example of this is the U.S-Korea Battery Cell Joint factories.
- Mandate the Department of Energy and the USITC to publish quarterly analyses of how tariffs affect LCOE for solar, storage, and EV battery prices. This analysis should support tariff adjustments and align them with deployment targets under the U.S. energy strategy
- Expand energy efficiency grants, such as EECBG grants, and revive programs, such as the Solar for All grants, to deliver residential solar and environmental justice for low-income households

Sensitivity Analysis

The sensitivity analysis tests the robustness of our preferred alternative, Differentiated Tariffs + Subsidies (Alternative 2), under changing policy assumptions and uncertain future conditions. Because stakeholder weights and impact scores may shift with changes in market dynamics, political priorities, or supply-chain disruptions, this section evaluates how variations in the utility model affect the final recommendation.

Our recommendation for a differential tariff strategy rests on current data that shows:

1) The U.S. has successfully diversified its imports for the solar sector away from China, relying primarily on Southeast Asian suppliers and rapidly expanding domestic manufacturing capacity, making higher solar tariffs both feasible and low-risk.

2) The battery sector in the U.S. remains heavily dependent on Chinese imports, particularly for lithium-ion batteries

3) By lowering battery tariffs while maintaining solar tariffs, the U.S. achieves both cost and equity benefits: EVs and storage become more affordable for consumers, especially low-income households.

Several unresolved uncertainties could shift the relative performance of the tariff alternatives evaluated in this analysis. First, the future pace of U.S. battery manufacturing expansion remains highly uncertain; accelerated domestic capacity growth could alter stakeholder weights toward employment and domestic sourcing, increasing the attractiveness of a high-tariff strategy. Second, the stability of diversified solar imports from Southeast Asian partners is not guaranteed. Trade investigations, geopolitical tensions, or supply chain disruptions could weaken the resilience of the solar sector and reduce the feasibility of maintaining elevated solar tariffs. Third, ongoing Congressional modifications to the Inflation Reduction Act introduce uncertainty about the long-term availability of manufacturing incentives, particularly the 45X and 48C credits. Faster-than-expected rollbacks could diminish the competitiveness of U.S. clean-energy manufacturing and shift the balance of utilities across alternatives. Fourth, potential Chinese retaliation, especially restrictions on critical minerals such as graphite, lithium compounds, or rare metals, could have broad cost implications across all pathways and undermine the assumptions of the utility model.

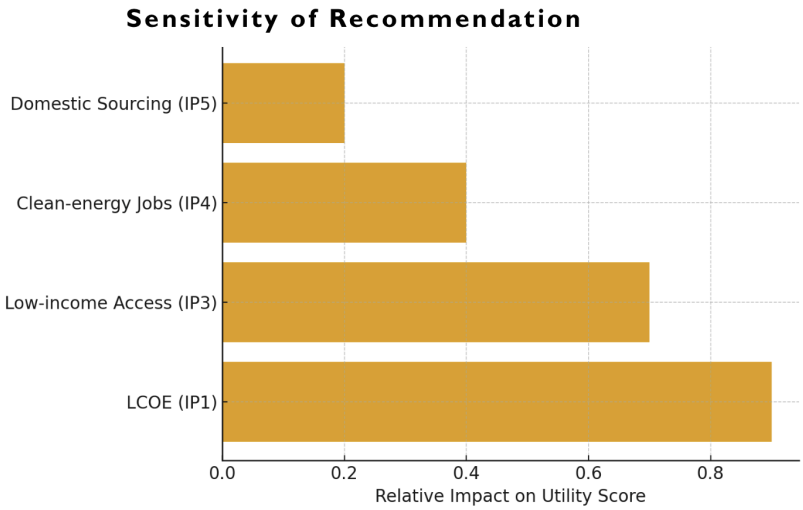
Based on these uncertainties, three scenarios are outlined below that quantify the range of validity of the indices of performance (IP)

1) National Security and Hard Industrial Policies (focus on geopolitics and rollback): In this scenario, IP4 and IP5 are prioritized, which are the number of clean energy jobs and domestic sourcing. IP1 and IP3, namely, LCOE and equity, would be deprioritized. In the utility model, Alternative 2 still dominates as long as IP1 and IP3 are reduced only by 50% and 1P4 and IP5 are increased by 0.25-0.30 across all stakeholders.

2) In case of severe supply chain disruptions due to geopolitical issues or tariff retaliation, IP5 would be prioritized, and in this case, Alternative 3 would overtake 2 only if all stakeholders weigh this factor> 0.85 across 4 stakeholder groups.

3) In a scenario where equity is expected to be handled outside trade policy, a natural assumption is that if policymakers rely on separate instruments such as EV rebates, income-based credits, or direct consumer subsidies, the weight placed on IP3 (low-income access) would fall sharply. As a result, Alternative 2 loses one of its major

advantages, since its strongest relative performance comes from lowering battery costs to improve affordability. In this context, tariff decisions become driven primarily by cost (IP1) and industrial policy objectives (IP4 and IP5), making Alternatives 1 or 3 more competitive depending on whether stakeholders prioritize deployment speed or domestic manufacturing strength.



The tornado diagram illustrates the relative influence of each (IP) on the overall utility score of the recommended policy alternative. It shows how sensitive the final decision is to changes in stakeholder weights across the four most consequential criteria: LCOE (IP1), Low-income Access (IP3), Clean-energy Jobs (IP4), and Domestic Sourcing (IP5). LCOE (IP1) exhibits the highest sensitivity. This indicates that even small changes in how stakeholders value cost impacts. Low-income Access (IP3) is the second most influential parameter. Alternative 2 gains significant relative utility from reducing battery costs for low-income households. Clean-energy Jobs (IP4) has a moderate influence. This reflects that Alternative 3 (high tariffs on both sectors) yields comparatively stronger results for domestic job creation. In scenarios where national security or industrial policy becomes the overriding priority, increased weight on IP4 could shift preference toward a more protectionist strategy.

Domestic Sourcing (IP5) has the lowest marginal effect among the four featured pathways, though it remains directionally important. Alternative 3 performs best on IP5, but large increases in the weight on domestic sourcing—approaching or exceeding 0.85 for several stakeholder groups, would be required before it overtakes Alternative 2. Under most plausible scenarios, IP5 variations alone do not overturn the preferred recommendation.

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